

Master Math for JEE Main & JEE Advanced

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JEE Mains 2020 Jan Chapter wise Question Bank

Nuclear Physics

Q1

On a photosensitive metal of area 1 cm^2 and work function 2eV , light of intensity $6.4 \times 10^{-5} \text{ W/cm}^2$ and wavelength 310 nm is incident normally. If 1 out of every 10^3 photons are successful, then number of photoelectrons emitted in one second is 10^x . Find x

7th Jan Morning

Sol

11

Energy of photon. $E = \frac{1240}{310} = 4\text{eV} > 2\text{eV}$ (so photoelectric effect will take place)

$$= 4 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-19} \text{ Joule}$$

No. of photons falling per second

$$= \frac{6.4 \times 10^{-5} \times 1}{6.4 \times 10^{-19}} = 10^{14}$$

No. of photoelectron emitted per second

$$= \frac{10^{14}}{10^3} = 10^{11}$$

Q2

Activity of a substance changes from 700 s^{-1} to 500 s^{-1} in 30 minute. Find its half-life in minutes

एक पदार्थ की सक्रियता 700 s^{-1} से 500 s^{-1} तक 30 मिनट में बदलती है, तब इसका अर्धआयुकाल मिनट में ज्ञात करो।

(1) 66

(2) 62

(3) 56

(4) 50

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(2)

$$\ln \left[\frac{A_0}{A_t} \right] = \lambda t$$

$$\Rightarrow \ln 2 = \lambda t_{1/2} \quad \dots(i)$$

$$\Rightarrow \ln \left[\frac{700}{500} \right] = \lambda (30 \text{ min}) \quad \dots(ii)$$

(i)/(ii)

$$\Rightarrow \frac{\ln 2}{\ln(7/5)} = \frac{t_{1/2}}{(30 \text{ min})}$$

$$\Rightarrow (2.06004) 30 = t_{1/2} = 61.8 \text{ min.}$$

Q3

Two photons of energy 4eV and 4.5eV are incident on two metals A and B respectively. Maximum kinetic energy for ejected electron is T_A for A and $T_B = T_A - 1.5\text{eV}$ for metal B. Relation between de-Broglie wavelengths of ejected electron of A and B are $\lambda_B = 2\lambda_A$. The work function of metal B is—

दो धातु A तथा B दी गई है। आपतित फोटोनों की ऊर्जा क्रमशः 4eV तथा 4.5eV है। उत्सर्जित इलेक्ट्रॉन की अधिकतम गतिज ऊर्जा धातु A के लिए T_A तथा धातु B के लिए $T_B = T_A - 1.5\text{eV}$ है। धातु A तथा B से उत्सर्जित इलेक्ट्रॉनों की डी-ब्रोगली तरंगदैर्घ्यों में सम्बन्ध $\lambda_B = 2\lambda_A$ है। धातु B के लिए कार्य फलन ज्ञात करो।

- (1) 3 eV (2) 1.5 eV (3) 4.5 eV (4) 4 eV

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Sol

(4)

Relation between De-Broglie wavelength and K.E. is

डी-ब्रोगली तरंगदैर्घ्य तथा गतिज ऊर्जा में सम्बन्ध

$$\lambda = \frac{h}{\sqrt{2(KE)m_e}} \Rightarrow \lambda \propto \frac{1}{\sqrt{KE}}$$

$$\frac{\lambda_A}{\lambda_B} = \frac{\sqrt{KE_B}}{\sqrt{KE_A}}$$

$$\Rightarrow \frac{1}{2} = \sqrt{\frac{T_A - 1.5}{T_A}} \Rightarrow T_A = 2 \text{ eV}$$

$$\therefore KE_B = 2 - 1.5 = 0.5 \text{ eV}$$

$$\phi_B = 4.5 - 0.5 = 4 \text{ eV}$$

Q4

An electron is moving initially with velocity $v_0\hat{i} + v_0\hat{j}$ in uniform electric field $\vec{E} = -E_0\hat{k}$. If initial wavelength of electron is λ_0 and mass of electron is m . Find wavelength of electron as a function of time.

एक इलेक्ट्रॉन प्रारम्भ में वेग $v_0\hat{i} + v_0\hat{j}$ से समरूप विद्युत क्षेत्र $\vec{E} = -E_0\hat{k}$ में गतिशील है। यदि इलेक्ट्रॉन की प्रारम्भिक तरंगदैर्घ्य λ_0 तथा द्रव्यमान m है तो इलेक्ट्रॉन की तरंगदैर्घ्य समय के पदों में ज्ञात करो।

$$(1) \frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2}{2m^2 v_0^2} t^2}}$$

$$(2) \frac{\lambda_0}{\sqrt{2 + \frac{e^2 E_0^2}{m^2 v_0^2} t^2}}$$

$$(3) \frac{\lambda_0 m v_0}{e E_0 t}$$

$$(4) \frac{2\lambda_0 m v_0}{e E_0 t}$$

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Sol

(1)

$$\text{Initially } m(\sqrt{2}v_0) = \frac{h}{\lambda_0}$$

$$\text{प्रारम्भ में } m(\sqrt{2}v_0) = \frac{h}{\lambda_0}$$

$$\text{Velocity as a function of time} = v_0\hat{i} + v_0\hat{j} + \frac{eE_0}{m}t\hat{k}$$

$$\text{समय के पदों में वेग} = v_0\hat{i} + v_0\hat{j} + \frac{eE_0}{m}t\hat{k}$$

$$\text{so wavelength } \lambda = \frac{h}{m\sqrt{2v_0^2 + \frac{e^2 E_0^2}{m^2} t^2}}$$

$$\text{अतः तरंगदैर्घ्य } \lambda = \frac{h}{m\sqrt{2v_0^2 + \frac{e^2 E_0^2}{m^2} t^2}}$$

$$\lambda = \frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2}{2m^2 v_0^2} t^2}}$$

Q5

Photons of wavelength $6556 \text{ \AA}$ falls on a metal surface. If ejected electrons with maximum K.E. moves in magnetic field of $3 \times 10^{-4} \text{ T}$ in circular orbit of radius 10^{-2} m , then work function of metal surface is

एक धात्विक सतह पर तरंगदैर्घ्य $6556 \text{ \AA}$ का एक फोटॉन आपतित है। यदि उत्सर्जित इलेक्ट्रॉन जिसकी अधिकतम गतिज ऊर्जा है, एक चुम्बकीय क्षेत्र $3 \times 10^{-4} \text{ T}$ में जिसकी त्रिज्या 10^{-2} m के वृत्ताकार कक्ष में गति करता है। तब धात्विक सतह का कार्यफलन ज्ञात करो –

- (1) 1.8 eV (2) 0.8 eV (3) 1.1 eV (4) 1.4 eV

9th Jan Morning

Sol

(3)

$$KE_{\max} = E - \phi$$

$$= \frac{12400}{\lambda(\text{in \AA})} - \phi \quad (\text{in eV})$$

$$\therefore r = \frac{\sqrt{2mKE}}{eB}$$

$$KE_{\max} = \frac{r^2 e^2 B^2}{2m} \quad (\text{in J})$$

$$= \frac{r^2 e B^2}{2m} \quad (\text{in eV})$$

$$\therefore \phi = \frac{12400}{6556} - \frac{r^2 e B^2}{2m}$$

$$= 1.1 \text{ eV}$$

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